Effective Low-cost Energy Saving Strategies in Supermarkets: An UK Case Study

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Abstract:
Supermarket buildings are complex energy systems that require careful study to make sure they perform in a sensible manner. The retail pressure of delivering stores in a short time makes engineering commissioning teams prone to mistakes and therefore, despite stores being newly built and carefully designed, their systems are not ideally set up; thus making the building underperform from day one. Consequently, energy savings are within easy reach if an effort is made to re-evaluate stores shortly after their opening date. This paper focuses on how adequate monitoring and good housekeeping can lead to effective energy saving strategies for a better management of services such as lighting, refrigeration, heating and ventilation. Additionally, a focused effort in curtailing energy use in supermarkets can also seriously reduce operational carbon related emissions; an ever-growing concern for retailers in an environment where sustainability practice is highly valued by consumers. A case study of a 35,000 ft2 supermarket located in the south-east of England serves as a vehicle to present and quantify effective, low-cost energy saving strategies. Extensive monitoring capabilities allow us to set a benchmark for all systems which then serves to assess the effectiveness of trials performed. Trials consist of: a) enhancing the dimming capabilities of the lighting system by improving sensor location and code, b) improving settings of fans and boiler system that reduce heating and ventilation requirements, and c) advocating the proper use of night-blinds in cabinets coupled with suction optimization of compressors that save energy use in the refrigeration system. All of these live trials have the feature of working closely with store staff and management, specialised contractors and academics. The synergy of parties allows the energy trials to succeed since aside from having solid technical foundations they have the full support from the people that work day in and day out in the supermarket – thus, neither commercial practices nor customer shopping experience is negatively affected by the energy trials. Overall, total store performance of the case study improved 15% against its benchmark, significantly reducing the carbon footprint and energy bills of the building. Furthermore, all the trials have quick payback periods with the virtue of being easy to roll-back and roll-forward across the estate; therefore implementation of such practices should be at the core of any energy efficiency program any food retailer embarks upon.

Keywords:
Energy efficiency, Monitoring based commissioning, Supermarkets, Sustainability.

1. Introduction

Supermarkets are one of the largest single-end users of electricity in the UK consuming approximately 3% of the UK's electricity and contributing to the UK's total GHG emissions by 1% [1]. Considering the challenge of Climate Change in conjunction with the UK's ambitious carbon emission reduction targets i.e. cutting GHG emissions by 80% below 1990 levels by 2050 [2], it becomes evident that energy conservation measures in supermarkets can substantially contribute to that direction.

However, the energy performance of the UK retailers has to be assessed in the context of their market. Total sales through UK grocery outlets were estimated at £156.8 billion in 2011, marking an increase of 3.8% compared to 2010 and are projected to be £182.7 billion by 2015 [3]. With 88,441 grocery stores in the UK in 2011 [3] and four major "players" in the market with more than 10% market share each (Tesco, ASDA, J Sainsbury, Wm Morrison) the industry is fiercely competitive [4]; moreover, their estate expansion plans imply that the competitive pressures will not decline. The competitive market share environment in conjunction with the tight operating margins

of UK food retailers (2% - 6% in 2004/05 according to [5]) increases the pressure for delivery of new stores in tight timelines. As a consequence, since the main focus is placed on the delivery of the store itself, the installation of the systems and their commissioning is prone to mistakes.

The term commissioning refers to the process of ensuring that building systems perform according to their design specifications and deliver the energy savings promised by their intended operation [6]. It also includes identifying energy saving opportunities for the existing equipment that does not include major capital investment on equipment [7].

Supermarket energy systems tend to underperform due to:

- Improper equipment installation, equipment degradation or incorrect control sequences [7];
- Meter and sensor related issues which can include sensor failure and/or meter connections even if the systems are operating properly indicating lack of commissioning and poor staff engagement [8].

Due to the factors above, commissioning is considered to be the single most cost-effective way to reduce energy use and as a result costs and GHG emissions in the built environment. However, even though commissioning practices gain in popularity the last years, they are still not utilized to their full potential [6].

The objective of this paper is to showcase the energy savings that can be achieved via low cost measures as well as the importance of monitoring-based commissioning in order to evaluate their effectiveness. The paper is organized as follows: Section 2 contains a brief description of the breakdown of energy use and carbon emissions in a supermarket as well as a technical approach to supermarket energy systems. Section 3 outlines the benefits of monitoring of energy use and indoor and ambient conditions in a supermarket. A description of the supermarket that was used for the case study is presented in Section 4 along with the results of the outcome of energy saving measures for the refrigeration, HVAC and lighting systems in Section 5. The paper closes with the conclusions drawn from this research in Section 6.

2. Energy systems in a supermarket

Approximately 70% of the energy used in UK supermarkets is in the form of electricity supplied from the grid to satisfy the needs for refrigeration, HVAC, lighting, baking and other auxiliary services. Natural gas compliments the remaining amount of energy demand in the form of space heating and hot water [1]. The breakdown of supermarket's electrical energy consumption to different end uses is as follows [9]:

- Refrigeration: 40% to 50%
- HVAC: 20% to 30%
- Lighting: 15% to 25%
- Hot water: 1% to 3%
- Other (Bakery, dry cleaners, rotisserie etc): 10% to 15%

The energy consumption in supermarkets has an adverse impact on the environment mainly in the form of greenhouse gas emissions (GHGs). Emissions due to energy use that is not generated on-site are classified as indirect. Electrical energy usage for the operation of the systems mentioned above falls commonly under this category.

On the other hand, direct emissions include those that are released in the atmosphere due to energy generation on-site or refrigerant gas leakage. The former can be mainly attributed to the use of

natural gas for thermal energy production for space heating and hot water purposes. For UK retailers, out of the total energy-related emissions, even though electrical energy consumption is around 70% of the total, it accounts for 80% to 90% of the energy-related emissions, the remainder being attributable to gas [10]. The reason for that is that carbon intensity of the UK's electricity grid is much higher than the combustion of natural gas (0.52037 $kgCO_2$/kWhel against 0.18521 $kgCO_2$/kWhth according to [11]).

Concerning the category of direct emissions, commercial refrigeration systems in supermarkets use large quantities of refrigerants that if leaked could be harmful to the atmosphere. This is because some of the substances used are potent GHGs, such as hydrofluorocarbons (HFCs), with high Global Warming Potential (GWP) which could be thousands of times greater than that of carbon dioxide (CO_2). Refrigerant losses from these systems add to the GHG emissions from these stores and in some cases they can be larger than indirect emissions [1]. According to [12], the emissions that can be associated to refrigerant leaks amount to around 15% of the energy-related emissions.

2.1. Refrigeration systems

The purpose of supermarket refrigeration systems is to keep perishable products refrigerated during storage and display times.

Typically in a supermarket there are two temperature levels of refrigeration: fresh food (dairy, meat etc.) is kept between 1°C and 14°C (medium temperature – MT or intermediate temperature – IT systems), whereas frozen food and ice creams are maintained at -12°C to -18°C (low temperature – LT systems) [13]. The corresponding evaporating temperature, for each system, depending on configuration, ranges from -15°C to -10°C and -40°C to -35°C, respectively. LT systems represent about 10% to 30% of installed capacity with while IT systems correspond to the remaining 70% to 90%. However, due to lower energy efficiency related to lower levels of temperature, LT systems consume 20% to 35% of the total refrigeration energy consumption [14]. There are three types of commercial refrigeration systems; these are [14]:

- Stand-alone systems: where the refrigeration system is integrated into the cabinet and heat is rejected into the store's sales area. These mainly are ice-cream freezers and other plug-in cabinets containing beer or soft drinks.
- Condensing units: are used mainly in convenience stores and consist of direct expansion systems that cool a small cold room or a small number of refrigerated cabinets.

Centralised system: used abundantly in the supermarket sector consists of a central refrigeration plant unit and its associated compressors in an enclosed room located either in the back or on the roof of the store. Heat is rejected via air-cooled condensers for least installation and maintenance costs or via evaporative condensers for less energy consumption due to lower condensing temperature [15]. This unit is used to provide display cabinets in the sales area as well as cold rooms in the back of the store with refrigerant that expands directly in air-refrigerant coils (DX – Direct Expansion system). Since these systems utilize hundreds of meters of piping to transport the refrigerant, they are prone to refrigerant leakage. Considering that commonly used refrigerants have Global Warming Potential (GWP) that is sometimes thousand times higher than CO_2, raises the need to carefully consider the technology to be installed.

In terms of refrigerant types, the majority of the installed systems utilize R404, while R134A, R407C and other hydrocarbons are also being extensively employed. To further reduce refrigerant charge and hence refrigerant leakage, secondary or indirect systems have been considered that employ natural refrigerants such as CO_2 (i.e. R744) and ammonia [16].

Finally, energy efficiency options for supermarket refrigeration systems include heat recovery systems to cover part or the whole of the heat demand of the store. Alternatively, float condensing schemes that allow the condensing temperature to fluctuate depending on ambient conditions, could significantly improve efficiency [17].

2.2. HVAC systems

Maintaining the indoor environment in supermarkets within the specified limits is essential for the comfort of customers and employees as well as to preserve the quality of the displayed products.

Supermarket HVAC systems pose certain challenges to designers, mainly due to the interaction between the internal environment and the cold air associated to large amounts of refrigeration fixtures. The refrigerated display cabinets remove considerable amounts of heat and moisture from the store; hence their effect has to be considered carefully in the design phase. Another challenge to the building designer is to make sure the store can remain positively pressurized during opening hours. The entrance and back doors of the store are usually "protected" with air curtains, automatic doors and/or lobby entrances to reduce outside air infiltration. However, the infiltration cannot be eliminated completely; hence it will always affect the heating/cooling and mechanical ventilation systems [1].

In a standard supermarket HVAC system, just like most commercial and retail environments, cooling is provided via a vapour compression chiller and heating via a natural gas boiler. Air Handling Units (AHUs) in conjunction with duct networks are employed for the supply of the heat or cooling to the different parts of the store and are usually installed on the roof or in the back of the store. Each unit consists of a fan and ducting for the supply, return and ventilation airflows. The heating and cooling coils are placed in the supply air duct [18].

High trading volumes from customers arriving and leaving the store, causing outside air infiltration, in conjunction with large amounts of open refrigerated cabinets in the sales area leads to space heating requirements being greater for supermarkets than other commercial buildings. Supermarkets in the UK usually also require cooling for a few days in the summer period. Therefore, some designs, instead of installing a dedicated chiller for space cooling purposes, employ a cold air retrieval system from the refrigerated cabinets which can be used to return the air to the air handling unit so that it can be re-circulated to other areas of the store – thus providing "free" cooling to the shop.

2.3. Lighting systems

Lighting is an important aspect of the store since there is a belief that bright light serves to highlight products and in general enhances the customer shopping experience [19].

The business as usual practices in supermarkets include the use of fluorescent lighting, T12 for older stores and T8 for newer ones [1]. Light emitting diodes (LEDs) is an alternative option that is being considered mainly due to projected energy savings [20-21] and their longer operating lives (50,000 to 100,000 hours) that can reduce maintenance costs. In the future, LED lighting is expected to become the norm; however, the main decision driver is expected to be trading performance and maintenance issues rather than energy efficiency [10].

The energy saving measures that are most prevalent in supermarkets include occupancy sensors, which can reduce energy consumption and maintenance costs by increasing the lifetime of the lighting fixtures, and automatic dimming controls combined with daylight systems to ensure minimum illumination levels are achieved while curtailing energy consumption [22]. Careful consideration is needed in the selection of the area of roof-lights for the store to minimize heat loss during the night and winter season as well as the risk of over-heating during the summer; u-values therefore need to be verified [10].

Finally, apart from lighting in the sales area, lighting inside the refrigerated display cabinets must meet aesthetic, functional and energy requirements. The vertical illuminance at any point within the display area must be at least 500 lux [23]. Lighting in cabinets constitutes approximately 25% of the energy consumption for refrigeration together with the evaporator fans [9] and it adds to the cooling load of the refrigerated cabinet. LED cabinets provide an even illuminance uniformity across the cabinet combined with a longer lifetime and up to 50% energy savings for cabinet lighting [9, 20]; convincing arguments for retailers to shift towards LEDs.

3. Energy monitoring and sub-metering in supermarkets

Metering of energy use is an essential part of the monitoring and targeting process that leads to effective energy management. The status quo for most commercial buildings is the installation of energy supply billing meters in order to identify a building's performance, for instance via a Display Energy Certificate. Nevertheless, an aggregated measure of energy consumption does not allow the determination of the reasons for bad performance or the areas where energy is wasted [8].

The sub-metering of building energy and water use allows the improved performance and operation of new and existing buildings; thus facilitating the activities of maintenance teams by providing clarity in the operation of the installed energy systems. Monitoring and recording data of past energy performance can also assist building operators in their effort to identify and design optimal and cost-effective future system designs and retrofit activities. From the building facilities manager perspective, sub-metering provides detailed information on energy consumption patterns that can help determine energy conservation measures. Finally, it can serve to evaluate the effectiveness of energy saving strategies applied either on systems or on behavioural changes [24].

In the UK, Part L2 of the Building Regulations [25-26] calls for energy meters so that the building operator can assign at least 90% of the estimated annual energy consumption of each fuel to the various end use categories (e.g. lighting, heating, ventilation etc.). In buildings with a total floor area greater than 1000 m2 the regulations require the enabling of automatic meter reading (AMR) and data logging. AMR systems provide building operators with an effective way of managing building energy use by accessing the data remotely in digital form, in a much more practical way than manual meter readings [8].

Careful consideration needs to be placed on the reading frequency of the meter. Reading every minute requires significantly more bandwidth and storage space compared to weekly readings. For the UK, the CRC (Carbon Reduction Commitment) schemes demands as a minimum half-hourly metered data [27]. Moreover, regular reporting of the energy use metered is essential to initiate actions that can save energy. Similarly to the reading frequency, too frequent reporting tends to be ignored completely, whereas too sparse reporting will not provide the necessary granularity to spot bad performance and energy waste [8]. Finally, installation of metering equipment that can be connected to BEMS (Building Energy Management Systems) systems for temporary data logging is a valuable way of investigating atypical energy consumption [28]. However, due to smaller storage capacity on BEMS systems, this option seems to be an ideal option for smaller buildings. In larger complex building, where a significant number of submeters are installed a dedicated AMR system can provide adequate data handling and storage capabilities [8].

In a supermarket context, as a minimum requirement, the systems that have to be sub-metered include the major energy consumers, as described in section 2, namely the refrigeration, HVAC and lighting system. Additionally, sub-meters could be placed in other areas of the store, such as the bakery and hot food ovens, and the employees and manager's area. In order to be able to get a better insight on the energy consumption of each different system though, it is not enough to measure the total energy consumption of each system. For instance, for the refrigeration system, it is strongly recommended that the energy consumption of the display cabinets and refrigeration packs are measured separately. Similarly for the HVAC system, measuring the energy demand of the AHUs as well as the boiler/chiller is essential to recognize where energy is consumed and design optimal energy saving strategies. Finally, for the lighting system, distinction between energy consumed in the sales area and energy for the car park area is considered to be the minimum requirement. Complementary to the aforementioned energy metering actions, measuring and logging data that will help analyze and get more insight on the energy consumption data should be a requirement. Knowing the variation of the temperature and humidity inside and outside the store will allow a better understanding of the energy use variation of the refrigeration and HVAC system. Moreover, the logging of lux levels values inside the store will facilitate the evaluation of the energy performance of the lighting system.

4. Supermarket case study

In this paper, the results of the energy saving strategies led by Imperial College and applied by contractors in a store in south-east of England are presented. The store has a sales area of 35,000 ft^2 or 3,300 m^2 and its opening hours are from 8am to 10pm on weekdays and Saturdays, and from 10am to 4pm on Sundays. Innovative building design concepts were utilised for this store and they include:

- Double glazed windows in the shop front;
- Reflective white ceilings and polished floor;
- Circular sun pipes on the ceiling;
- Extensive side windows in the general merchandise area.

Using the monitoring system installed in the supermarket, it has been possible to gather energy data for the first year of the store's operation, namely from February 2011 to February 2012. This period corresponds to the time after commissioning of the systems, but before any energy saving measures were implemented. The breakdown of energy use of the store in the different subsystems is shown in Table 1.

Table 1. Benchmarking supermarket energy use for the first year of operation.

End use	**Electricity (kWh)**	**Share (%)**
Refrigeration	747,207	42
Lighting	522,234	29
HVAC	191,377	11
Bakery & Hot Food	254,150	14
Other	77,963	4
Total	1,792,931	100
	Heat (kWh)	
Boiler	905,698	

This strategy allows the targeting of the most energy intensive areas of the store and assists in the design of effective energy saving strategies. Most importantly, though, when analysed in half-hourly, daily or weekly form; this data can be used in order to create benchmarks both for the store's total electricity use as well for its different subsystems in order to evaluate the effectiveness of any energy saving measures.

4.1. Energy trials and results

In this case study, a monitoring-based commissioning approach was adopted that can be seen in Fig. 1 [7]. The term refers to a cyclical process that starts by gathering building information via mechanical drawings, equipment specifications and energy monitoring systems. Benchmarking, the second step in the process, allows the comparison between the examined building's energy performance and the evaluation of energy saving measures that will be taken in the following steps. Fault detection is a key step in the process, and according to [7] it can be performed using trend-data analyses, functional testing, and/or fault detection and diagnostic tools. In the fourth step, the energy saving measures that have been identified by the commissioning engineer are implemented and finally energy savings are assessed using actual energy consumption data [7].

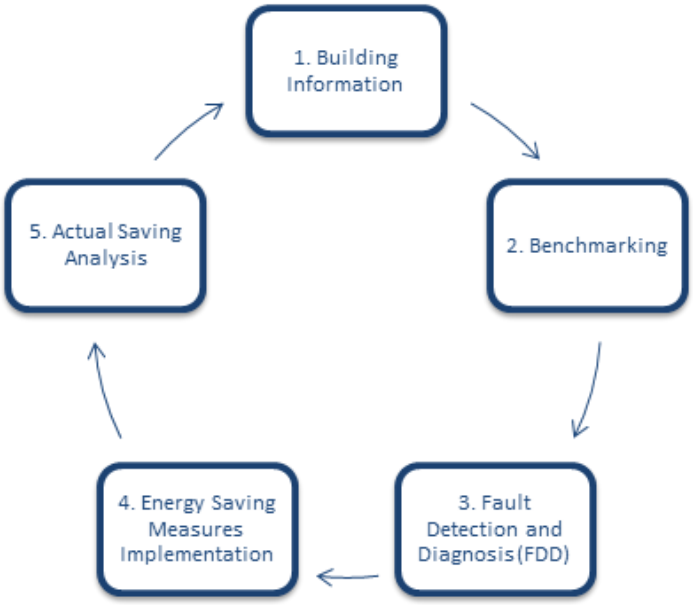

Fig. 1. Monitoring based commissioning flow chart [7].

4.2. Refrigeration results

The refrigeration system, being the major consumer of electricity in a supermarket, has been the system where the majority of efforts were focused due to its great energy consumption. The energy saving strategies that target the refrigeration system are described below:

- Temperature setpoint drop: The first energy saving measure that was investigated is not directly related to the refrigeration system, but it indirectly positively influenced its energy consumption. Changing the ambient temperature setpoint of the supermarket from 19˚C to 18˚C during trading hours and from 19˚C to 16˚C during non-trading hours has had a positive effect on the refrigeration system, by decreasing the temperature of the warm air that infiltrates the refrigerator cabinets on the sales floor.
- Cabinet night blinds: Even though doors on refrigerator cabinets can have a significant impact on the system's energy consumption, while also improving the ambient climate of the supermarkets [29], the perception that it might negatively affect the turnover of the store, has prevented the implementation of such a scheme in the studied store. Instead, an alternative approach was taken that involved the use of night blinds during non-trading hours to reduce warm air infiltration losses.

  The use of night blinds during non-trading hours, though, affects the usual activities that take place inside the store during these times which are mainly replenishing the shelves of the cabinets with products. In order to ensure, that the store's employees would comply with the use of the night blinds, the authors worked in collaboration with the store manager, firstly to observe and map the activities inside the sales area after the closing of the store and then to design an optimal replenishment strategy with the minimum number of night blinds being pulled up.

  To ensure the continuous utilisation of the installed night blinds, a weekly report of night blind usage is sent to the store manager. Monitoring the refrigeration's system energy consumption every 30 minutes, it is possible through the energy profiles to understand whether the night blinds were pulled down and up in accordance with the store's trading hours. If the night blinds are pulled on time, then the energy profile shows a steep energy decrease or increase for the closing and opening time respectively; on the other hand, if the night blinds were not pulled down on schedule, the energy consumption profile shows a less steep gradient.

Figures 2 and 3 show respectively, how best practice looks like of night-blind usage and the type of weekly scorecard sent to management to inform them of their performance.

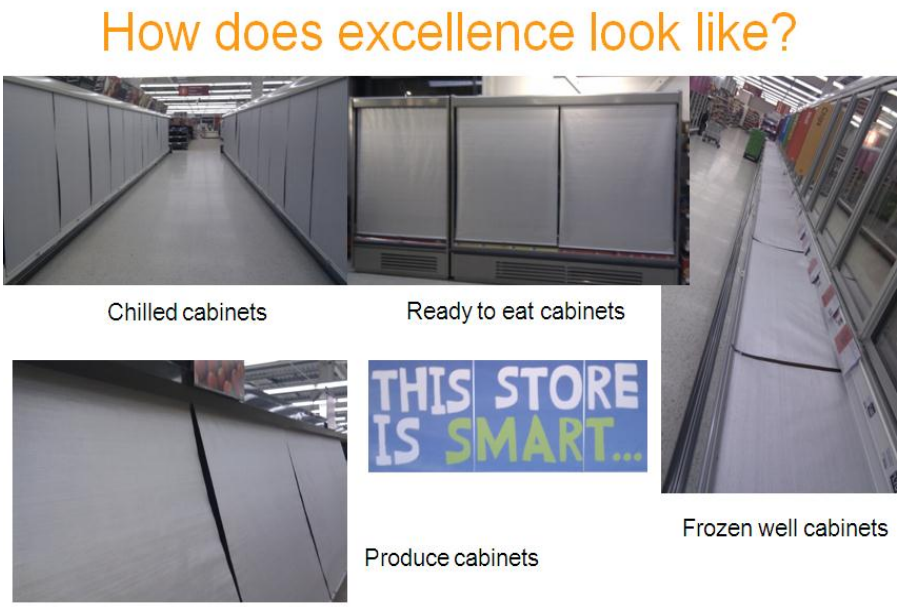

Fig. 2. Creating energy awareness with colleagues is essential to save energy on refrigeration demand – messages such as these highlight the effort to make night-blind use effective.

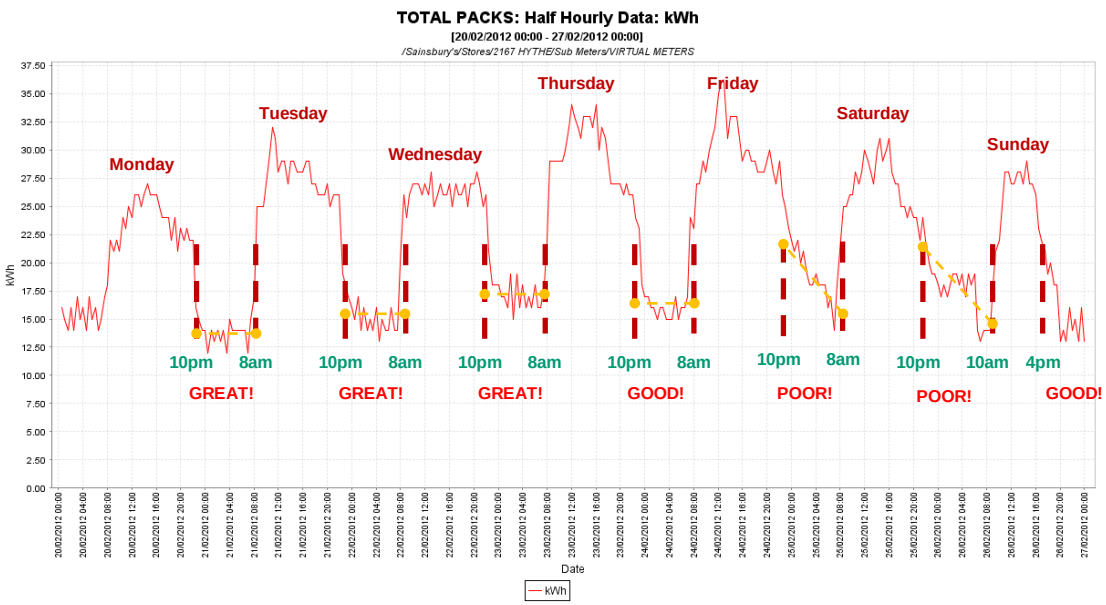

Fig. 3. Data capture of refrigeration pack demand allows us to visualise if night-blinds are being used effectively.

- Suction Pressure Optimisation: a more technical approach was taken to further reduce the system's energy demand. Working closely with the store's refrigeration controls contractor a suction pressure optimisation scheme was introduced. Monitoring the temperature of each evaporator and depending on the deviation from its setpoint, it is possible to understand whether the evaporator is cold enough or not. If the setpoints are satisfied, the controller is allowed to raise the suction pressures that it controls within limits set by its optimizer, hence reducing the energy consumption of compressor packs.

4.3. HVAC results

In this section, enhancements of the operation of the HVAC and boiler system are described. In this specific case, any of the energy saving measures that were considered, were always evaluated from a customer thermal comfort perspective. The reason is that even though energy consumption reduction was the main goal of the analysis, careful consideration was needed to not affect the sense of comfort for customers that could adversely impact the store shopping and trading activities. The energy saving measures that were applied on HVAC are described below:

- Variable Speed Drive (VSD) settings: In a conventional HVAC system fans are designed to operate at constant speed. In order to follow the heating and/or cooling load some form of mechanical throttling is used to reduce water or air flow in the system. However, the system fans still operate at full speed consuming the same amount of energy regardless of the load. If a VSD scheme is utilized, the fans have the capability to speed-up or slow down following demands. Since motor energy use is a function of speed, energy savings are generated through operation at less than full load. Taking into account that systems are oversized to account for extreme future loads the energy savings become even more significant. In the examined store the fans of the AHUs' fans were given a lower and a maximum operating speed, since these were commissioned at fixed speeds, hence leading to significant energy consumption reduction.
- Occupational/Non-occupational setpoint drop: The first measure that directly aimed to the reduction of the boiler's energy consumption was the modification of the occupational and non-occupational setpoints of the store. By reducing the space temperature setpoint from 19°C for the whole day to 18°C and 16°C during trading and non-trading hours respectively, substantial boiler savings can be achieved without adversely affecting the thermal comfort of the building occupants.
- HVAC flow setpoint drop: Another simple measure that was implemented is the change of the setpoint of the hot water flow in the AHUs of the store from 85°C to 82°C. This change aims to decrease the energy consumption of the boiler by decreasing the temperature that the water is supplied to the AHU.
- Hot Water System (HWS) setpoint drop: Similarly to the HVAC flow setpoint drop, the setpoint of the HWS system was dropped from 70°C to 65°C to further reduce the energy consumption of the boiler

4.4. Lighting results

Ideal lighting must provide proper levels of illumination for the desired activity to be performed inside a building with a minimum use of energy. Under this premise, research was conducted to optimise the programming code that controlled the dimming light system; this work has been published in detail before [30]. The strategies that were implemented are described below:

- Changing lux level setpoints: Taking advantage of the digital dimmable technology being used by the ballasts, a detailed curve explaining the relationship between lux levels and power use was determined. The ability to identify power reduction potential as a function of lux output allowed us to conduct several trials that gradually reduced light intensity. After a few months of engaging with stakeholders, it was confirmed no negative trade impact had resulted from the trials and the business decided to officially drop its specification from 900 lux to 650 lux while trading and from 250 lux to 200 lux during non-trading. During the first phase of the trials, management was asked to say whether they had noticed that lighting levels of the store had been reduced and if their job had been affected. They mentioned that they had barely realized the change; however both visual comfort as well as their ability to conduct their routine activities was not affected.
- Relocation of sensor: By monitoring the energy use of the lighting system in conjunction with natural daylight levels inside the store, it became evident that the lighting system did not dim as effectively as the technology allowed it to. After conducting a site visit, the

authors were able to determine that some light sensors were misplaced during the commissioning of the system. By working with the system's contractor the sensor was relocated and the system is now performing according to its specifications

4.5. Total store results

In this section, the results of the energy consumption of the store, after the implementation of the energy saving measures discussed previously are presented. The results as seen on Table 2 represent the stores systems performance for the period of February 2012 to February 2013; savings on the refrigeration, HVAC (with heating), and lighting services are considerable and highlight the substantial benefit low-cost energy saving strategies can be achieved post-commissioning. Nonetheless, energy was not possible to be reduced in the bakery & hot food area due to extended cooking times in the 2nd year of store operations. The "Other" loads was another category were energy was not possible to be reduced; consisting of office areas, staff kitchen and other ancillary services which unfortunately are not thoroughly sub-metered.

Table 2. Supermarket energy use during its second year of operation.

End use	Electricity (kWh)	Change (%)
Refrigeration	649,056	-13.1%
Lighting	470,940	-9.8%
HVAC	131,131.72	-31.5%
Bakery & Hot Food	267,209	+5%
Other	80,383	+3%
Total	1,594,049	-12.6%
	Heat (kWh)	
Boiler	631,289	-30.3%

Figure 4 details and compares how each energy system in the supermarket performed on a 'like for like basis'; showing the substantial improvement the energy trials have provided. Total electricity demand was down from 1,792 MWh to 1,594 – 12.6% reduction. Similarly, heating demand provided by the boiler was reduced from 905 MWh to 631 MWh; a 30% drop despite the 2nd year being colder and thus with a higher heating degree days metric.

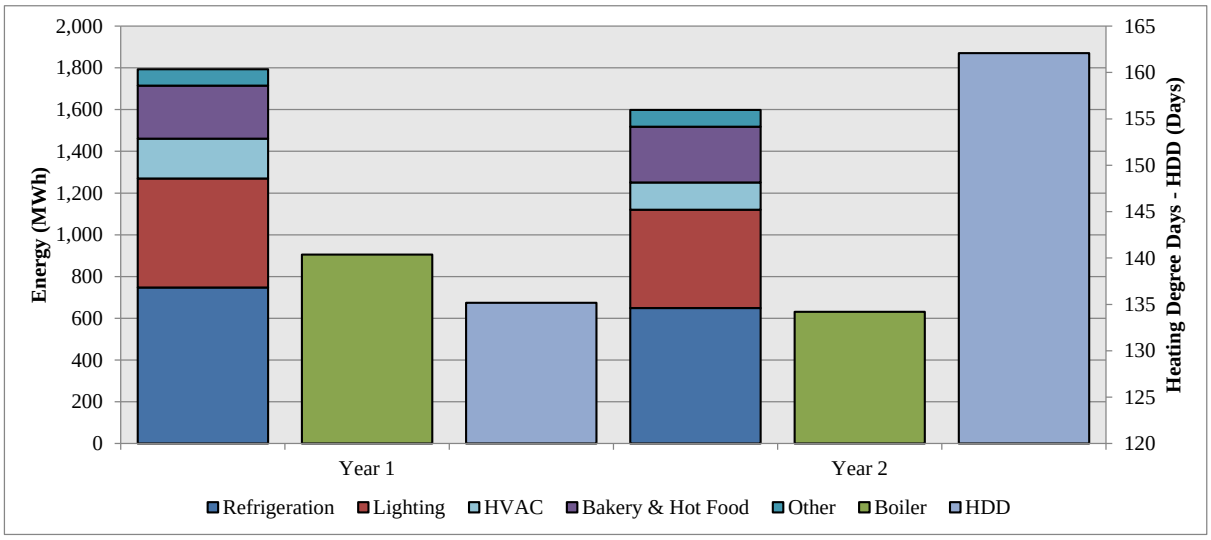

Fig. 4. 'Like for like' comparison between energy store systems – year 1 vs. year 2.

Despite the attractive numbers in reducing absolute kilo-watt hours consumed energy and environmental engineers are also focused on reducing CO_2 emissions and operating costs. Table 3 shows that since the store is heating using a biomass boiler annual savings on CO_2 were only 1.7 tonnes, nevertheless the carbon reduction on the electricity side was an attractive 144 tonnes. Under the CRC scheme being proposed by the UK government, at £12 tCO_2, the emissions saved during the 2nd year of operation would equate to about £1,752 in additional expenses. Meanwhile, savings on the electricity bill amounted to almost £20,000. As the results can attest, the sound practices of running buildings as best as they can is a rewarding effort that empowers colleagues in stores and enhances the green credentials of the business, while also helping to reduce unnecessary operating costs and mitigating carbon emissions.

Table 3. Supermarket performance for the first year and second years of operation.

Time Period	**Electricity – OPEX (£)**	**Heat – OPEX (£)**	**Electricity CO_2 emissions (tCO_2)**	**Heat CO_2 emissions (tCO_2)**
Year 1	157,381	50,162	988	5.43
Year 2	137,566	34,964	864	3.79
Change (%)	-12.6%	-30.3%	-12.6%	-30.3%

5. Conclusions

Supermarket buildings are complex energy systems that operate under a highly competitive environment where the customer is always right; therefore energy efficiency measures should not compromise shopping experience. This paper focuses on how adequate monitoring and good housekeeping can lead to effective low-cost energy savings for a better management of services. Particular emphasis has been given to reviewing the purpose of lighting, refrigeration, and HVAC systems, while also describing their usual energy demands. After careful research on the specifications and control capabilities of each system, sensible energy trials were devised and applied with the support of contractors and store management. These trials had the benefit of being conducted once sufficient energy benchmarking data had been collected; thus allowing us to evaluate impacts. Store performance was monitored on a weekly basis to make sure benefits were maintained and to report any unexpected performance to relevant stakeholders. The holistic approach in this work has shown that with adequate monitoring and analytics capabilities savings in electricity and heat are possible; 12% and 30% respectively. These benefits have also important positive consequences in carbon emissions and operating costs. If scaled across thousands of supermarkets these energy strategies could yield multi-million annual savings and increased benefits to businesses as it would also translate into reducing their CRC costs and improving their bottom line. Best of all is that these energy initiatives are quite straight-forward to replicate. Rolling-back across the estate these learning's may seem a challenging task but with adequate resources and a robust project plan the benefits should be achievable with quick economic returns. Further work in this research area should focus on creating robust and reliable tools that can automate intelligent operation and reporting of commercial building performance.

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